



Practice Paper A

Multiple choice question book

Chemistry

Paper 1



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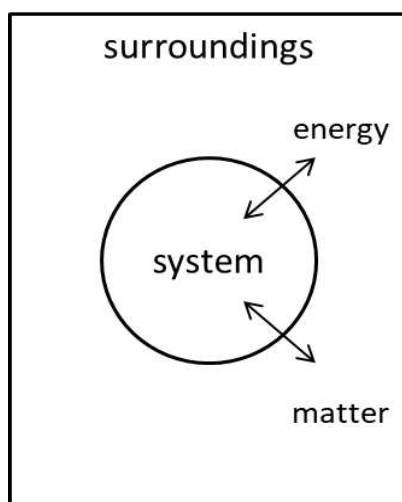
Section 1

Instructions

- Answer all questions in the question and response book.
 - This book will not be marked.
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QUESTION 1

The figure below demonstrates which type of system?



- (A) open
- (B) closed
- (C) isolated
- (D) homogenous

QUESTION 2

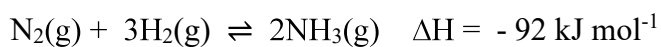
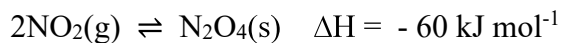
What scientific concept is defined by the statement,

“for a reaction to occur, the particles must collide with sufficient energy to break the bonds and have the appropriate orientation to allow the new bonds to form”?

- (A) collision theory
- (B) static equilibrium
- (C) dynamic equilibrium
- (D) Le Chatelier’s principle

QUESTION 3

Consider the two reactions at equilibrium shown below:

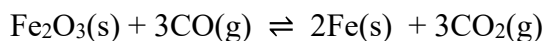


Which action will drive BOTH reactions to form more product?

- (A) reducing the pressure
- (B) increasing the reaction temperature
- (C) increasing the volume of the reaction vessel
- (D) removing the product from the reaction vessel

QUESTION 4

Which alternative most correctly represents the equilibrium expression for the following reaction?



- (A) $\frac{[\text{CO}_2]^3}{[\text{CO}]^3}$
- (B) $\frac{[\text{CO}_2]^3}{[\text{Fe}_2\text{O}_3][\text{CO}]^3}$
- (C) $\frac{[\text{Fe}_2\text{O}_3][\text{CO}]^3}{[\text{Fe}]^2[\text{CO}]^3}$
- (D) $\frac{[\text{Fe}]^2[\text{CO}_2]^3}{[\text{Fe}_2\text{O}_3][\text{CO}]^3}$

QUESTION 5

What is the pH of a 0.05 M solution of potassium hydroxide?

- (A) 1.3
- (B) 12.7
- (C) 13
- (D) 5×10^{-2}

QUESTION 6

Deduce the conjugate acid of SO_4^{2-} .

- (A) SO_4^{2-}
- (B) HSO_4^-
- (C) H_2SO_4
- (D) HSO_4^{2-}

QUESTION 7

The table shows the colours of three indicators as the hydrogen ion concentration changes.

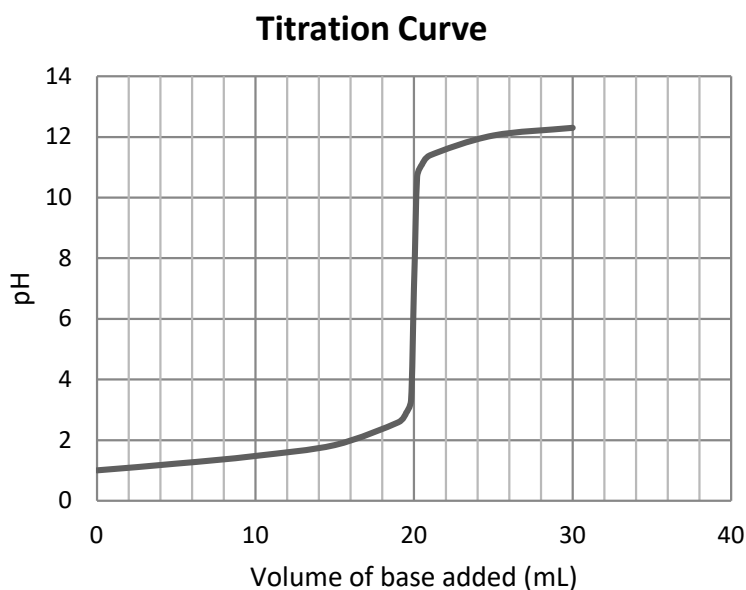
$[\text{H}^+]$ mol L ⁻¹	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
<i>Methyl orange</i>	red	orange	yellow	yellow	yellow
<i>Bromothymol blue</i>	yellow	yellow	green	blue	blue
<i>Phenolphthalein</i>	colourless	colourless	colourless	colourless	pink

A solution is colourless when tested with phenolphthalein, green with bromothymol blue and yellow when tested with methyl orange.

Which statement about the solution is correct?

- (A) The solution is weakly acidic with a pH of 6.
- (B) The solution is strongly acidic with a pH of 6.
- (C) The solution is weakly alkaline with a pH of 8.
- (D) The solution is strongly alkaline with a pH of 8.

Use the following graph to answer Questions 8 and 9.



QUESTION 8

Which statement about this titration is correct?

- (A) The acid was placed in the burette and the equivalence point occurred at pH 7.
- (B) The base was placed in the burette and the equivalence point occurred at pH 12.
- (C) The acid was placed in the conical flask and the equivalence point occurred at pH 7.
- (D) The acid was placed in the conical flask and the equivalence point occurred at pH 12.

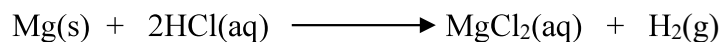
QUESTION 9

Which pair of solutions would be most likely to produce this titration curve?

- (A) citric acid and sodium hydroxide
- (B) acetic acid and ammonia solution
- (C) sulfuric acid and ammonia solution
- (D) hydrochloric acid and sodium hydroxide

QUESTION 10

Consider the following reaction



Which statement is correct?

- (A) H_2 is reduced.
- (B) Cl^- is reduced.
- (C) Mg is oxidised.
- (D) Mg is reduced.

QUESTION 11

What is the oxidation number of Cr in $\text{Cr}_2\text{O}_7^{2-}$?

- (A) +2
- (B) +3
- (C) +5
- (D) +6

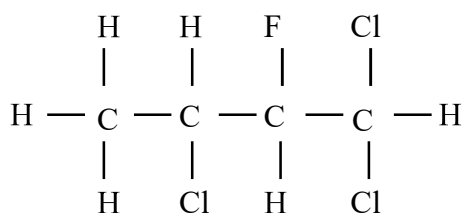
QUESTION 12

Deduce how many isomers are there for $\text{C}_2\text{H}_2\text{Br}_2\text{Cl}_2$.

- (A) two
- (B) three
- (C) four
- (D) five

QUESTION 13

The figure represents the structural formula for a molecule.



Deduce the correct name for this haloalkane.

- (A) 1,1-dichloro-2-fluoro-methylpropane
- (B) 1,1,3-trichloro-2-fluoro-butane
- (C) 2,4,4-trichloro-3-fluoro-butane
- (D) 2-fluoro-1,1,3-trichloro-methylpropane

QUESTION 14

The table lists some boiling point temperatures of a range of organic molecules.

<i>Carboxylic acid</i>	<i>BP (°C)</i>	<i>Alcohol</i>	<i>BP (°C)</i>
methanoic	100.6	methanol	64.5
ethanoic	117.9	ethanol	78.3
propanoic	140.8	1-propanol	97.2

What is the main reason why carboxylic acids have a higher boiling point than alcohols?

- (A) Alcohols have a higher molecular weight than carboxylic acids.
- (B) Carboxylic acids form strong dispersion forces between molecules.
- (C) Carboxylic acids have strong ionic bonds between the acid molecules.
- (D) Carboxylic acids have stronger hydrogen bonding compared to alcohols.

QUESTION 15

A liquid rapidly decolourises bromine water after shaking.

Which liquid is most likely to react in this way?

- (A) heptane
- (B) 2-heptene
- (C) 1-heptanol
- (D) heptanoic acid

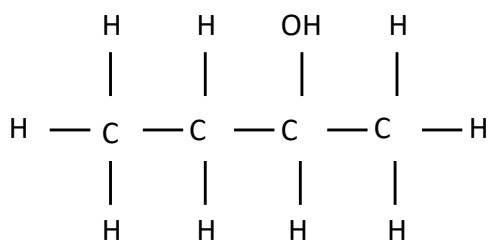
QUESTION 16

Which equation represents esterification?

- (A) $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) \rightarrow 2\text{C}_2\text{H}_5\text{OH}(\text{aq}) + 2\text{CO}_2(\text{g})$
- (B) $\text{CH}_3\text{COOH}(\text{l}) + \text{C}_2\text{H}_5\text{OH}(\text{l}) \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5(\text{l}) + \text{H}_2\text{O}(\text{l})$
- (C) $\text{CH}_3\text{COOH}(\text{l}) + \text{KOH}(\text{aq}) \rightarrow \text{CH}_3\text{COONa}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- (D) $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) + \text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) \rightarrow \text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

QUESTION 17

Acidified potassium permanganate solution is a powerful oxidising agent. It is added to a substance with the following structural formula;

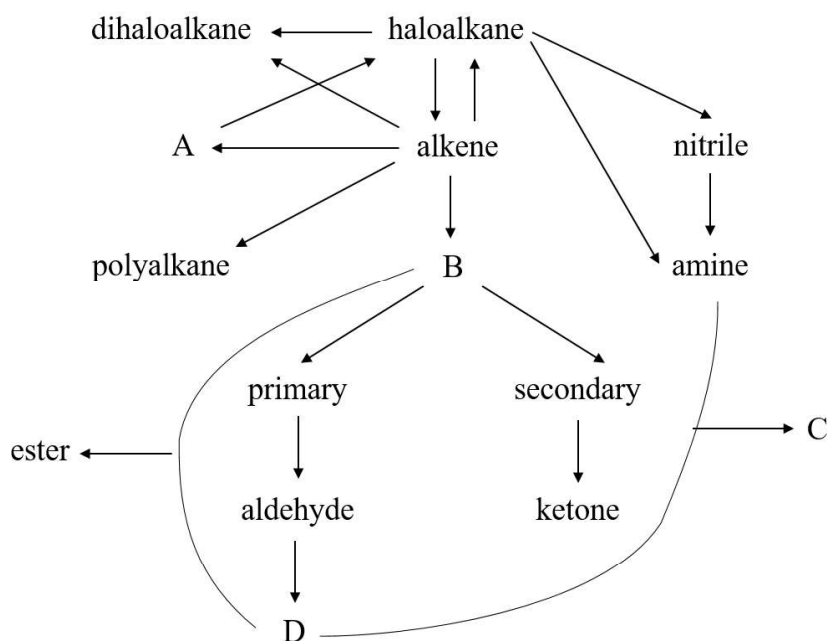


What is the name of the organic product of this reaction?

- (A) butanol
- (B) butanal
- (C) butanone
- (D) butanoic acid

QUESTION 18

The figure below shows several common chemical synthesis pathways.



Determine the type of organic compound most likely represented by D.

- (A) amide
- (B) alkane
- (C) alcohol
- (D) carboxylic acid

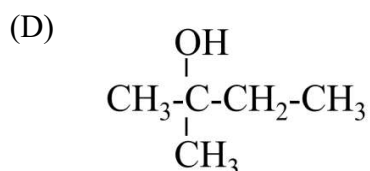
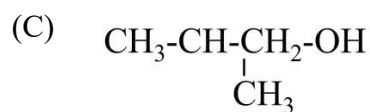
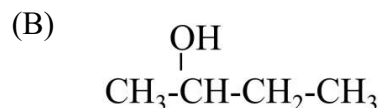
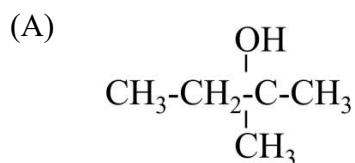
QUESTION 19

Monosaccharides are organic molecules that contain either

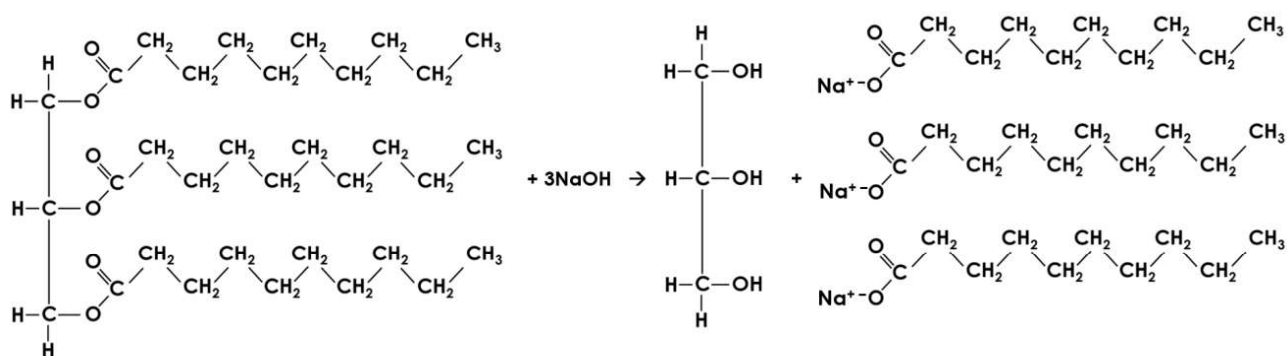
- (A) a ketone group or an amine group and several hydroxyl groups.
- (B) a carbonyl group or an ester group and several hydroxyl groups.
- (C) an aldehyde group or a ketone group and several hydroxyl groups.
- (D) a carboxyl group or an aldehyde group and several hydroxyl groups.

QUESTION 20

Which of the structural formulae below is that of a secondary alcohol?

**QUESTION 21**

The structural equation below demonstrates how glycerol and fatty acid salts can be formed by reacting a triglyceride with a base.



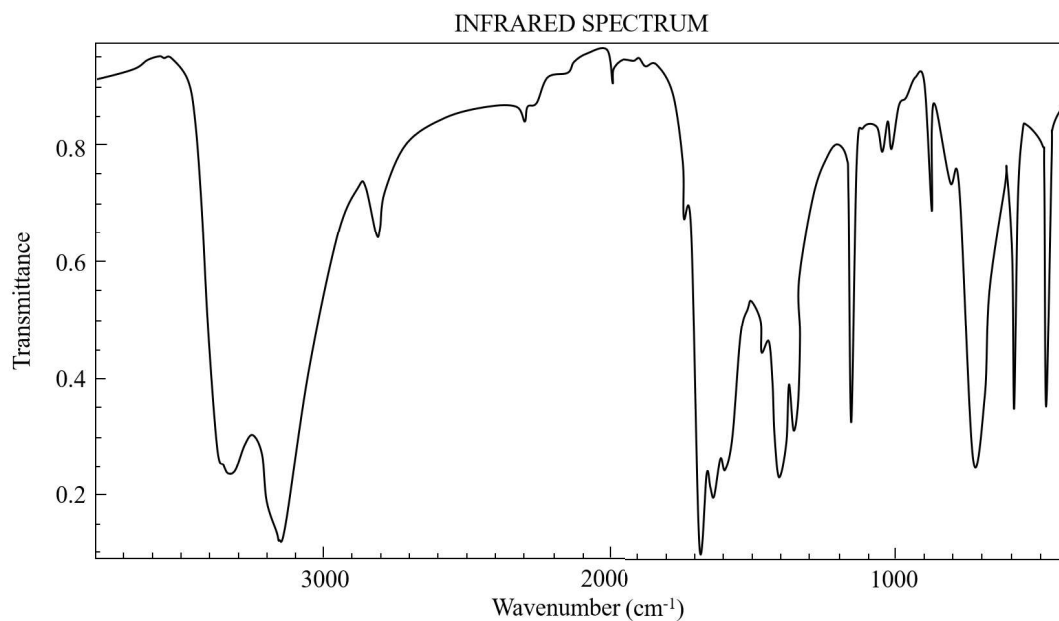
What is the name given to this type of reaction?

- (A) halogenation
- (B) esterification
- (C) saponification
- (D) hydrogenation

QUESTION 22

An unknown sample was analysed using infrared spectroscopy.

What is the compound?



- (A) ethene
- (B) ethanol
- (C) ethanamide
- (D) ethanoic acid

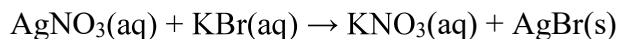
QUESTION 23

Determine the set of conditions required for the production of ethanol by fermentation.

- (A) anaerobic, 35°C, yeast, glucose solution
- (B) anaerobic, 20°C, yeast, glucose solution
- (C) aerobic, 15°C, dilute sulfuric acid, starch
- (D) aerobic, 30°C, dilute sulfuric acid, starch

QUESTION 24

A precipitate of silver bromide, AgBr, can be produced by reacting solutions of silver nitrate and potassium bromide:



If A_r of Ag = 108, A_r of N = 14, A_r of O = 16, A_r of K = 39 and A_r of Br = 80, the atom economy for this reaction would be closest to:

- (A) 34.9%
- (B) 41.2%
- (C) 58.8%
- (D) 65.1%

QUESTION 25

Identify which of these molecules is a monomer used in the production of an addition polymer.

- (A) CH₄
- (B) C₂H₆
- (C) C₂H₃Cl
- (D) C₃H₇Cl